```
010D 2AF008             LHLD    HSAVE1  ;IF "X" KEY IS NOT PRESSED
0110 77                 MOV     M,A     ;DATA MAY BE REENTER FOR
0111 21FD08     REDO:   LXI     H,DSPBFA; PRESENT LOCATION.
0114 22EC08             SHLD    HLDISP
0117 2AF008             LHLD    HSAVE1
011A 23                 INX     H
011B 22F008             SHLD    HSAVE1
011E 7C                 MOV     A,H

011F CD2F01             CALL    HLFORM
0122 2AF008             LHLD    HSAVE1  ;ADDRESS AND DATA ENTERED
0125 7D                 MOV     A,L     ;IS DISPLAYED ON LED'S
0126 CD2F01             CALL    HLFORM
0129 2AF008             LHLD    HSAVE1
012C C3F800             JMP     NEXT
012F 32FE08     HLFORM: STA     DIGIT   ;THIS SUBROUTINE WILL
0132 E6F0               ANI     0F0H    ;FORMAT A BINARY CHARACTOR
0134 0F                 RRC             ;IN REG. A TO A DISPLAY
0135 0F                 RRC             ;CHARACTOR AND THEN STORE IT
0136 0F                 RRC             ;IN THE DISPLAY BUFFER.
0137 0F                 RRC
0138 CD4D02             CALL    CONHEX  ;
013B 3AFE08             LDA     DIGIT
013E E60F               ANI     0FH
0140 CD4D02             CALL    CONHEX
0143 C9                 RET
0144 22F008     DSDATA: SHLD    HSAVE1  ;THE DATA POINTED TO BY H&L
0147 7E                 MOV     A,M     ;IS FOEMATED
0148 0F                 RRC             ;BY THIS SUBROUTINE AND
0149 0F                 RRC             ;STORED IN THE DISPLAY BUFFER
014A 0F                 RRC
014B 0F                 RRC
014C E60F               ANI     0FH
014E CD4D02             CALL    CONHEX
0151 2AF008             LHLD    HSAVE1
0154 7E                 MOV     A,M
0155 E60F               ANI     0FH
0157 CD4D02             CALL    CONHEX
015A C9                 RET
                ;
                ;SCAN KEYPAD & REFRESH DISPLAY
                ;
015B 0E07       SCAN:   MVI     C,07H   ;SET UP TO SCAN KEYPAD
015D 0D         SCAN1:  DCR     C
015E CA5B01             JZ      SCAN
0161 CD6A01             CALL    SCAN2   ;SCAN ONE COLUMN
0164 C28F01             JNZ     SCAN4   ;IF KEY PRESSED GO DEBOUNCE KEY,
0167 C35D01             JMP     SCAN1   ;IF NOT, SCAN NEXT COLUMN
016A 79         SCAN2:  MOV     A,C     ;SCANING COLUMN AND OUTPUTING
016B 3D                 DCR     A       ;DATA FROM DISPLAY BUFFER
016C D300               OUT     DRA     ;TO LED DISPLAY
016E 11F808             LXI     D,DSPBFD
0171 83                 ADD     E
0172 5F                 MOV     E,A
0173 1A                 LDAX    D
0174 D301               OUT     DRB
0176 CD8101             CALL    DELAY
0179 DB00       SCAN3:  IN      DRA     ;CHECK FOR KEY PRESS
017B E6F0               ANI     0F0H
017D 2F                 CMA
017E E6F0               ANI     0F0H
0180 C9                 RET
0181 110420     DELAY:  LXI     D,2004H ;DEBOUNCE DELAY & DISPLAY
0184 AF         TIME:   XRA     A       ;REFRESHTIME
0185 1D                 DCR     E       ;5MS DELAY
0186 C8                 RZ
0187 15         LOOP:   DCR     D
0188 BA                 CMP     D
0189 CA8401             JZ      TIME
018C C38701             JMP     LOOP
018F 3E00       SCAN4:  MVI     A,0H    ;DEBOUNCE SUBROUTINE
0191 D301               OUT     DRB
0193 CDF004             CALL    KDELAY
0196 CDF004             CALL    KDELAY
0199 CD7901             CALL    SCAN3
019C CA5B01             JZ      SCAN
019F DB00       DECODE: IN      DRA     ;READ KEYPAD & DECODE
01A1 4F                 MOV     C,A
01A2 CD7901     TEST:   CALL    SCAN3   ;WAIT UNTIL KEY IS RELEASED
01A5 C2A201             JNZ     TEST
01A8 CD8101             CALL    DELAY
01AB CD8101             CALL    DELAY
01AE 79                 MOV     A,C     ;THEN DECODE THE KEY PRESSED
```

```
01AF 21B903              LXI     H,INTBL
01B2 0E18                MVI     C,018H
01B4 BE          DCODE1: CMP     M       ;COMPARE KEY PRESSED WITH
01B5 CAC201              JZ      CHAR    ;INPUT TABLE
01B8 23                  INX     H
01B9 23                  INX     H
01BA 23                  INX     H
01BB 0D                  DCR     C
01BC CA5B01              JZ      SCAN    ;KEY PRESSED NOT IN TABLE
01BF C3B401              JMP     DCODE1
01C2 2B          CHAR:   DCX     H       ;GET HEX CHARACTER FROM TABLE
01C3 2B                  DCX     H
01C4 7E                  MOV     A,M
01C5 32FE08              STA     DIGIT
01C8 C9                  RET
                 ;
                 ;INPUT CHARACTERS FORM KEYPAD
                 ;
01C9 21FD08      INPUT4: LXI     H,DSPBFA;INPUT FOUR CHARACTER
01CC 0604                MVI     B,04H
01CE CDFC01              CALL    INPUTI
01D1 CD0A02      INPUTA: CALL    INPUTC
01D4 FE10                CPI     X       ;IF ANY COMMAND KEY PRESSED
01D6 D28A00              JNC     COMMD   ;GOTO COMMAND ROUTINE
01D9 05                  DCR     B
01DA C2D101              JNZ     INPUTA
01DD C32A02              JMP     INFROM
01E0 21F908      INPUT2: LXI     H,DSPBFD+1
01E3 0602                MVI     B,02H
01E5 CDFC01              CALL    INPUTI
01E8 CD0A02      INPUTB: CALL    INPUTC
01EB FE10                CPI     X       ;IF "X" IS PRESSED SET CARRY
01ED CAFA01              JZ      SETCY   ;AND RETURN
01F0 D28A00              JNC     COMMD   ;ANY OTHER COMMAND KEY JUMP
01F3 05                  DCR     B       ;TO COMMAND ROUTINE
01F4 C2E801              JNZ     INPUTB
01F7 C32A02              JMP     INFROM
01FA 37          SETCY:  STC
01FB C9                  RET
01FC 22EC08      INPUTI: SHLD    HLDISP  ;SET UP DISPLAY AND INPUT
01FF 21F308              LXI     H,INBUFF; BUFFERS
0202 22EE08              SHLD    HSAVE2
0205 78                  MOV     A,B
0206 32EB08              STA     DGTCNT
0209 C9                  RET
020A CD5B01      INPUTC: CALL    SCAN
020D F5                  PUSH    PSW
020E 78                  MOV     A,B     ;CLEAR LED ON NEW ENTRY
020F FE02                CPI     2
0211 C21A02              JNZ     CONTIN
0214 210000              LXI     H,0
0217 22F808              SHLD    DSPBFD
021A F1          CONTIN: POP     PSW
021B 2AEE08              LHLD    HSAVE2  ;INPUT & DISPLAY ENTERED DIGITS
021E 77                  MOV     M,A     ;CONVERT HEX TO DISPLAY
021F 23                  INX     H       ;CHARACTERS
0220 22EE08              SHLD    HSAVE2
0223 CD4D02              CALL    CONHEX
0226 3AFE08              LDA     DIGIT
0229 C9                  RET
022A 21F308      INFROM: LXI     H,INBUFF; FORMAT SUBROUTINE
022D 3AEB08              LDA     DGTCNT  ; THIS SUBROUTINE WILL CONVERT
0230 D601                SUI     01H     ; DATA ENTER FROM KEYPAD.
0232 47                  MOV     B,A     ; FOUR KEY STROKES INTO
0233 7E                  MOV     A,M     ; TWO BYTES OR TWO KEY
0234 07                  RLC             ; STROKES INTO ONE BYTE
0235 07                  RLC
0236 07                  RLC
0237 07                  RLC
0238 23                  INX     H
0239 86                  ADD     M
023A 05                  DCR     B       ; REG. B CONTAINES THE # OF
023B C8                  RZ              ; KEY STROKES
023C 77                  MOV     M,A
023D 23                  INX     H
023E 7E                  MOV     A,M
023F 07                  RLC
0240 07                  RLC
0241 07                  RLC
0242 07                  RLC
0243 23                  INX     H
0244 86                  ADD     M
0245 2B                  DCX     H
```

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0246 2B              DCX     H
0247 2B              DCX     H
0248 77              MOV     M,A
0249 2AF308          LHLD    INBUFF
024C C9              RET
024D 21B703    CONHEX: LXI   H,HEXTB ;THIS SUBROUTINE CONVERT
0250 BE        CNHEX1: CMP   M       ;HEX TO A DISPLAY CHARACTER
0251 CA5A02            JZ    CNHEX2  ;AND STORES IT IN THE
0254 23                INX   H       ;DISPLAY BUFFER
0255 23                INX   H
0256 23                INX   H
0257 C35002            JMP   CNHEX1
025A 23        CNHEX2: INX   H
025B 7E                MOV   A,M
025C 2AEC08            LHLD  HLDISP
025F 77                MOV   M,A
0260 2B                DCX   H
0261 22EC08            SHLD  HLDISP
0264 C9                RET
                ;
                ;EXECUTE ROUTINE
                ;
0265 CD6902    EXCUTE: CALL  ENADDR
0268 E9                PCHL
0269 CDD902    ENADDR: CALL  ENTER   ;ENTER FOUR CHARACTERS VIA
026C CDC901            CALL  INPUT4  ;KEYPAD, AND STORE IT IN H&L.
026F E5                PUSH  H       ;THEN TRANSFER H&L TO PC
0270 CD5B01            CALL  SCAN
0273 FE10              CPI   X
0275 C28A00            JNZ   COMMD   ;JUMP TO COMMAND ROUTINE
0278 E1                POP   H       ;IF ANY COMMAND KEY OTHER
0279 C9                RET           ;THEN "X" IS PRESSED.
                ;
                ;SET BREAK POINT ROUTINE
                ;
027A CDDF00    BRKPT:  CALL  MOVE1   ;MOVE # OF BREAK POINTS
027D 21E208            LXI   H,NOBKRG; PREVIOUSLY SET TO DISPLAY
0280 7E                MOV   A,M
0281 47                MOV   B,A     ;IF THREE HAVE BEEN SET GO
0282 EB                XCHG          ;DISPLAY OVER ERROR
0283 21F908            LXI   H,DSPBFD+1
0286 22EC08            SHLD  HLDISP
0289 CD4D02            CALL  CONHEX
028C 214040            LXI   H,4040H
028F 6F                MOV   L,A
0290 22F808            SHLD  DSPBFD
0293 78                MOV   A,B
0294 3C                INR   A
0295 FE04              CPI   04H
0297 D2CE02            JNC   OVERR
029A EB                XCHG
029B 85                ADD   L
029C 6F                MOV   L,A
029D E5                PUSH  H
029E CDC901            CALL  INPUT4
02A1 7E                MOV   A,M     ;PUT RST 1 AT LOCATION
02A2 EB                XCHG          ;SPECIFIED BY USER AND STORE
02A3 E1                POP   H       ;USED'S INSTRUCTION IN BREAK
02A4 77                MOV   M,A     ;POINT BUFFER
02A5 EB                XCHG
02A6 3ECF              MVI   A,RST1
02A8 77                MOV   M,A
02A9 3E79              MVI   A,079H
02AB 32F908            STA   DSPBFD+1
02AE 3E54              MVI   A,054H
02B0 32F808            STA   DSPBFD
02B3 E5                PUSH  H
02B4 D5                PUSH  D
02B5 CD5B01            CALL  SCAN
02B8 FE10              CPI   X
02BA CAC402            JZ    INCRMT
02BD E1                POP   H
02BE 7E                MOV   A,M
02BF E1                POP   H
02C0 77                MOV   M,A
02C1 C37A02            JMP   BRKPT
02C4 3AE208    INCRMT: LDA   NOBKRG  ;INCREMENT # OF BREAK POINTS
02C7 3C                INR   A
02C8 32E208            STA   NOBKRG  ;THEN RETURN TO COMMAND ROUTINE
02CB C38A00            JMP   COMMD
02CE 0606      OVERR:  MVI   B,6     ;SUBROUTINE TO OUTPUT OVER
02D0 11F808            LXI   D,DSPBFD; ERROR WHEN MORE THEN THREE
02D3 21B103            LXI   H,OVERMG; BREAK POINTS ARE ENTER.
```

```
02D6 C3CA00             JMP     ERROR1
02D9 214040     ENTER:  LXI     H,4040H ;ADDRESS PROMPT
02DC 22F808             SHLD    DSPBFD
02DF C9                 RET
                ;
                ;DISPLAY CPU REGISTER ROUTINE
                ;
02E0 F5         DISREG: PUSH    PSW     ;SUBROUTINE TO SAVE ALL
02E1 3AFF08             LDA     USRCSR  ;CPU REGISTER.
02E4 E63F               ANI     03FH
02E6 F640               ORI     040H
02E8 D308               OUT     CSR1    ;TURN OFF 8155'S TIMER
02EA C5                 PUSH    B
02EB D5                 PUSH    D
02EC E5                 PUSH    H
02ED 21FCFF             LXI     H,0FFFCH;SAVE STACK POINTER
02F0 39                 DAD     SP      ;POINTING TO WHAT IT WOULD
02F1 E5                 PUSH    H       ;BE AT END OF STACK
02F2 EB                 XCHG
02F3 20                 DB      20H     ;"RIM INSTRUCTION"
02F4 F5                 PUSH    PSW
02F5 210C00             LXI     H,000CH ;POINT H&L TO PROGRAM COUNTER
02F8 39                 DAD     SP      ;IN STACK
02F9 22EE08             SHLD    HSAVE2
02FC CD0A03             CALL    DREG    ;GO DISPLAY CPU REGISTER
02FF 3AE108             LDA     INTERF  ;IF IN SINGLE STEP MODE
0302 FE00               CPI     0H      ;JUMP TO STIMER
0304 C27903             JNZ     STIMER
0307 C31B00             JMP     RESREG  ;IF NOT GO RESTORE REGISTER
030A EB         DREG:   XCHG            ;DISPLAY CPU REG. SUBROUTINE
030B AF                 XRA     A
030C 77                 MOV     M,A
030D EB                 XCHG
030E 3E07               MVI     A,07H
0310 F5                 PUSH    PSW
0311 21FD08             LXI     H,DSPBFA
0314 22EC08             SHLD    HLDISP
0317 2AEE08             LHLD    HSAVE2
031A 23                 INX     H
031B CD4401             CALL    DSDATA
031E 2AF008             LHLD    HSAVE1
0321 2B                 DCX     H
0322 22EE08             SHLD    HSAVE2
0325 CD4401             CALL    DSDATA
0328 219503             LXI     H,NAMETB-2
032B F1                 POP     PSW
032C F5                 PUSH    PSW
032D 87                 ADD     A
032E 85                 ADD     L
032F 6F                 MOV     L,A
0330 3E00               MVI     A,0H
0332 8C                 ADC     H
0333 67                 MOV     H,A
0334 11F808             LXI     D,DSPBFD
0337 0602               MVI     B,02H
0339 CDE700             CALL    MOVE
033C CD5B01             CALL    SCAN
033F FE15               CPI     S
0341 CA5703             JZ      SINSTP-2
0344 FE10               CPI     X
0346 C28A00             JNZ     COMMD
0349 F1                 POP     PSW
034A 3D                 DCR     A
034B C8                 RZ
034C 2AEE08             LHLD    HSAVE2
034F 2B                 DCX     H
0350 2B                 DCX     H
0351 22EE08             SHLD    HSAVE2
0354 C31003             JMP     DREG+6
0357 F1                 POP     PSW
0358 C9                 RET
                ;
                ;SINGLE STEP ROUTINE
                ;
0359 3E80       SINSTP: MVI     A,080H  ;STEP THUR USER'S PROGRAM
035B 32E108             STA     INTERF  ;FROM ADDRESS ENTER
035E CD6902             CALL    ENADDR
0361 3E00               MVI     A,0H
0363 D30D               OUT     TIMERH
0365 3E5A               MVI     A,05AH  ;SET UP 8155'S TIMER
0367 D30C               OUT     TIMERL
0369 3AFF08             LDA     USRCSR  ;LOAD USER'S IMAGE OF 8155'S
036C F6C0               ORI     0C0H    ;CSR REG. AND START TIMER
```

```
036E D308                  OUT     CSR1
0370 3E05       DELAY2     MVI     A,05H    ;DELAY SAME AMOUNT OF TIME
0372 3D                    DCR     A        ;THAT IT WOULD TAKE TO
0373 C27203                JNZ     DELAY2+2; RESTORE CPU REGISTER
0376 00                    NOP
0377 00                    NOP
0378 E9                    PCHL             ;GOTO USER'S PROGRAM
0379 3E00       STIMER:    MVI     A,0H     ;ENTER HERE TO SET 8155'S
037B D30D                  OUT     TIMERH   ;TIMER WHEN STEP THRU USER'S
037D 3E5A                  MVI     A,05AH   ;PROGRAMAND RESTORE USER'S
037F D30C                  OUT     TIMERL   ;CPU REGISTER
0381 3AFF08                LDA     USRCSR
0384 F6C0                  ORI     0C0H     ;START TIMER
0386 D308                  OUT     CSR1
0388 C31B00                JMP     RESREG
038B C3E002FF00UMOVE:      DB      0C3H,0E0H,02H,0FFH,0H,0H
0391 FF0000FF00            DB      0FFH,0H,0H,0FFH,0H,0H
0397 D0B0F3EDB8NAMETB:     DB      0D0H,0B0H,0F3H,0EDH,0B8H,0F6H,0F9H,0DEH
039F B9FCF1F7B9            DB      0B9H,0FCH,0F1H,0F7H,0B9H,0F3H
03A5 0000000000SINEIN      DB      0H,0H,0H,0H,0H,040H
03AB 0000005050ERRMAG:     DB      0H,0H,0H,050H,050H,079H
03B1 4040505079OVERMG:     DB      040H,040H,050H,050H,079H,03FH
03B7 00         HEXTB:     DB      0H       ;ENTER HERE FOR HEX TABLE
03B8 3F         DISPTB:    DB      03FH     ;ENTER HERE FOR DISPLAY TABLE
03B9 E1         INTBL:     DB      0E1H     ;ENTER HERE FOR INPUT TABLE
03BA 0106E2025B            DB      01H,06H,0E2H,02H,05BH,0D2H,03H,04FH
03C2 B204667205            DB      0B2H,04H,066H,072H,05H,06DH,071H,06H
03CA 7DB10707D1            DB      07DH,0B1H,07H,07H,0D1H,08H,07FH,070H
03D2 0967B00A77            DB      09H,067H,0B0H,0AH,077H,0D0H,0BH,07CH
03DA E00C397D0D            DB      0E0H,0CH,039H,07DH,0DH,05EH,0BDH,0EH
03E2 79DD0F71ED            DB      079H,0DDH,0FH,071H,0EDH,010H,040H,0ECH
03EA 1173BC1238            DB      011H,073H,0BCH,012H,038H,0DCH,013H,03EH
03F2 7C1476EB15            DB      07CH,014H,076H,0EBH,015H,030H,0DBH,016H
03FA 00BB17507B            DB      0H,0BBH,017H,050H,07BH
03FF 3830383520            DB      '8085 MONITOR V7.15 COPYRIGHT'
041B 2843292031            DB      '(C) 1982 BELL & HOWELL'
0431 4544554341            DB      'EDUCATION GROUP, INC'
0445 414C4C2052            DB      'ALL RIGHTS RESERVED'
                           ;
                           ;THE FOLLOWING IS A 15MS DELAY
                           ;
                           ;
04F0                       ORG     BEGIN+04F0H
04F0 E5         KDELAY:    PUSH    H
04F1 F5                    PUSH    PSW
04F2 210007                LXI     H,0700H
04F5 2B         KD1:       DCX     H
04F6 7C                    MOV     A,H
04F7 B5                    ORA     L
04F8 C2F504                JNZ     KD1
04FB F1                    POP     PSW
04FC E1                    POP     H
04FD C9                    RET
                           ;
                           ;
                           ;MEMORY & I/O TEST
                           ;FOR THE MM-85 KIT
                           ;
                           ;VECTOR LOCATION IN MON8085 V7.10
                           ;
0000 =                     MONITOR EQU     0000H
                           ;
                           ;MEMORY TEST PROGRAM STARTS AT 0500 HEX
                           ;
0500                       ORG     BEGIN+0500H
0500 210014     MEMTEST:   LXI     H,1400H
0503 1604                  MVI     D,04H
0505 0E00       NEXTA:     MVI     C,00H
0507 3E00                  MVI     A,00H
0509 C31105                JMP     START1
050C 3C         NEXT1:     INR     A
050D 0D                    DCR     C
050E CA1E05                JZ      HERE
0511 77         START1:    MOV     M,A
0512 47                    MOV     B,A
0513 7E                    MOV     A,M
0514 B8                    CMP     B
0515 CA0C05                JZ      NEXT1
0518 CDE002                CALL    DISREG
051B C33005                JMP     LEDTEST
051E 23         HERE:      INX     H
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051F AF                 XRA     A
0520 BD                 CMP     L
0521 C20505             JNZ     NEXTA
0524 15                 DCR     D
0525 C20505             JNZ     NEXTA
0528 C33005             JMP     LEDTEST
                        ;
                        ;I/O TEST PROGRAM STARTS AT 0530
                        ;
0530                    ORG     BEGIN+0530H
0530 CDDF00     LEDTEST:CALL    MOVE1
0533 CDD902             CALL    ENTER
0536 CD5B01             CALL    SCAN
0539 3E0F               MVI     A,0FH
053B D308               OUT     08H
053D 3EAA       HERE1:  MVI     A,0AAH
053F D309               OUT     09H
0541 D30A               OUT     0AH
0543 D30B               OUT     0BH
0545 CD5B01             CALL    SCAN
0548 3E55               MVI     A,055H
054A D309               OUT     09H
054C D30A               OUT     0AH
054E D30B               OUT     0BH
0550 CD5B01             CALL    SCAN
0553 C33D05             JMP     HERE1
                        ;
                        ;
                        ;
                        ; THE FOLLOWING SUBROUTINES SUBMITTED BY B. KONIKKARA
                        ;      OHIO INSTITUDE OF TECHNOLOGY....
                        ;
                        ;
                        ;THIS SUBROUTINE WILL DISPLAY CONTENTS OF ACCUMULATOR
                        ;IN THE DATA FIELD FOR APPROXIMATELY ON SECOND, THEN
                        ;CLEAR DISPLAY AND RETURN.  ALL REGS. SAVED
                        ;ARE RESTORED.
                        ;
                        ;
0560                    ORG     BEGIN+0560H
0560 F5         DISPLAY:PUSH    PSW
0561 C5                 PUSH    B
0562 D5                 PUSH    D
0563 E5                 PUSH    H
0564 CDB205             CALL    CLEAR   ;CLEARS DISPLAY BUFFER
                        ;                SEPARATES ACC INTO LED
                        ;                FORM. REG. B GETS HIGH
                        ;                NIBBLE. C, LOW NIBBLE
0567 CD8B05             CALL    SEPATE  ;DISPLAY TIME LOOP COUNTER
056A 2680               MVI     H,080H  ;HIGH NIBBLE INTO ACC
056C 78         BACK:   MOV     A,B     ;GET READY TO DISPLAY
056D D301               OUT     DRB     ;SECOND LED
056F 3E01               MVI     A,1
0571 D300               OUT     DRA
0573 CD8101             CALL    DELAY   ;LET IT BE SEEN!
0576 79                 MOV     A,C     ;LOW NIBBLE
0577 D301               OUT     DRB
0579 97                 SUB     A
057A D300               OUT     DRA     ;DISPLA LOW NIBBLE
057C CD8101             CALL    DELAY
057F 25                 DCR     H
0580 C26C05             JNZ     BACK    ;DO IT 80H TIMES
0583 97                 SUB     A
0584 D301               OUT     DRB     ;CLEAR DISPLAY
0586 E1                 POP     H
0587 D1                 POP     D
0588 C1                 POP     B
0589 F1                 POP     PSW     ;RESTORE STUFF
058A C9                 RET
                        ;
                        ;
                        ;       SEPARATE SUBROUTINE
                        ;        ACC INTO B AND C
                        ;            LED FORM
                        ;
                        ;
058B F5         SEPATE: PUSH    PSW
058C 47                 MOV     B,A
058D E60F               ANI     0FH     ;CLEAR LEFT NIBBLE
058F CD4D02             CALL    CONHEX  ;GET LED FORM
0592 4F                 MOV     C,A     ;SAVE RIGHT LED NIBBLE IN C
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0593 78                 MOV     A,B       ;GET ORIGINAL #
0594 E6F0               ANI     0F0H      ;CLEAR RIGHT NIBBLE
0596 0F                 RRC
0597 0F                 RRC
0598 0F                 RRC               ;MOVE LEFT NIBBLE
0599 0F                 RRC               ;INTO RIGHT NIBBLE
059A CD4D02             CALL    CONHEX    ;CONVERT TO LED FROM
059D 47                 MOV     B,A       ;SAVE IN REG B.
059E F1                 POP     PSW
059F C9                 RET
                ;
                ;
                ;       GET SUBROUTINE
                ;
                ;WHEN CALLED DISPLAY A DECIMAL POINT ON THE LEFT
                ;MOST LED AND WAITS UNTIL ANY KEY IS PRESSED.
                ;RETURNS KEY VALUE IN ACC. USES SCAN ROUTINE.
                ;
                ;
05A0 C5         GET:    PUSH    B
05A1 D5                 PUSH    D
05A2 E5                 PUSH    H
05A3 CDB205             CALL    CLEAR
05A6 3E80               MVI     A,080H    ;MOVE DECIMAL POINT
05A8 32FD08             STA     DSPBFA    ;TO LAST LED IN DISPLAY.
05AB CD5B01             CALL    SCAN      ;GO GET KEY
05AE E1                 POP     H
05AF D1                 POP     D
05B0 C1                 POP     B
05B1 C9                 RET
                ;
                ;       CLEAR SUBROUTINE
                ;       CLEANS UP DISPLAY BUFFER AREA
                ;
05B2 E5         CLEAR:  PUSH    H
05B3 210000             LXI     H,0
05B6 22F808             SHLD    DSPBFD
05B9 22FA08             SHLD    DSPBFD+2
05BC 22FC08             SHLD    DSPBFD+4
05BF E1                 POP     H
05C0 C9                 RET
                ;
                ;
                ;       FIVE00 SUBROUTINE IS JUST
                ;       A 500 MICRO SEC. DELAY!!
                ;
                ;
05C1 F5         FIVE00: PUSH    PSW
05C2 3E60               MVI     A,060H    ;COUNTER
05C4 3D         AGAIN:  DCR     A
05C5 C2C405             JNZ     AGAIN
05C8 F1                 POP     PSW
05C9 C9                 RET
                ;
                ;
                ;       "BEEP"  BEEPS FOR APPROXIMATELY HALF
                ;               A SECOND AT 1 KHZ ON S.O.D. PIN.
                ;
                ;
05D0                    ORG     BEGIN+05D0H

05D0 F5         BEEP:   PUSH    PSW
05D1 E5                 PUSH    H
05D2 2670               MVI     H,070H
05D4 3EC0       TONE:   MVI     A,0C0H    ;SOD = 1
05D6 30                 DB      30H       ;"SIM   INSTRUCTION"
05D7 CDC105             CALL    FIVE00
05DA 3E40               MVI     A,040H    ;SOD    = 0
05DC 30                 DB      30H
05DD CDC105             CALL    FIVE00
05E0 25                 DCR     H
05E1 C2D405             JNZ     TONE
05E4 E1                 POP     H
05E5 F1                 POP     PSW
05E6 C9                 RET
05E7                    END
05E7
006H USE FACTOR
END OF ASSEMBLY
```